

ActInf Livestream #015.0 ~ "Free-Energy Principle, Computationalism and

Livestream | 1:36:16

hello everyone welcome to
the active inference lab the active
inference live stream
today it is active inference live stream
15.0
and it is january 27 2021
welcome to the active inference lab
everyone we are an experiment in
online team communication learning and
practice related to
active inference you can find us at our
website
twitter email youtube key based team
also with the discord now
and this is recorded in an archived live
stream so please provide us with
feedback
so that we can improve on our work all
backgrounds and perspectives are
welcome here and this is going to be fun
because it's a dialogue which i'm not
sure if it's a format we've tried before
but we'll abide by
good video chat etiquette with blue and
i so just to introduce ourselves before
we jump into everything
i'm daniel friedman i'm a postdoctoral
researcher
in davis california and i'm an insect
biologist mostly what about yourself
blue
my name is blue knight i am an
independent research consultant based
out of
new mexico and my background is mostly

in neuroscience

cool well this is going to be a fun

conversation

these conversations are related to the

active inference live stream

group discussions that we do regularly

and these discussions are every tuesday

from 7

to 9 a.m pacific and if you go to this

red link you'll find a spreadsheet and

the spreadsheet shows the two

participatory group discussions that we

spend the time on

with each paper and so for february 2nd

and 9th

2021 for active stream 15.1 and 15.2

we're going to be talking about this

paper that we're going to discuss

in 15.0 with a bigger group as well as

with the authors in s

and thomas so this is going to be a cool

opportunity to hear a lot of

perspectives of people who are just

learning

and also have the authors on for some

long form discussions

with time to digest so the dot zero

videos

um which is what this video is has the

goal of setting the context for these

group discussions

because these are research papers and

they're

basically the form of the ideas as

they're being worked with by researchers

so it is something that might take a lot

of background reading or thinking about

to catch up with

so the goal is maybe these videos in the
dot zero

can prepare someone to feel more
comfortable participating in
point one or point two even

participating as a listener

so the paper we're going to be
discussing for today and for the next
two weeks

is called free energy principle
computationalism

and realism a tragedy it's by thomas van
ness and in s

hippolyto and the link is provided here
the video is just an introduction to the
context of the ideas it's not a review
or a final word especially if you knew
how much time we spent preparing for
this

and the punchline of the paper if we
could put it

this way is that realism is a tragedy
and instrumentalism is no joke either
the sections of today are going to go as
follows in the introduction

we're going to go through the aims and
the claims of the papers

then the keywords the abstract and the
roadmap

then there's one figure one table and a
few

key topics and quotations that really
stand out

as far as summarizing some of the bigger
ideas that we've been talking about

but also really crystallizing the
arguments of the paper so good things to
catch on to if you didn't highlight it

or draw a circle around it when you were reading it earlier and then we'll close with a couple questions that we had just while making the slides

and these aren't ones that we're going to address today but they're things that we're curious to hear what people think about and what other questions and methods they connect them out to so in 15.1 and 0.2 we're going to be discussing

this very same paper so just submit your questions through the live chat or through a youtube comment or twitter however and then if you want to um participate live in these or a future discussion then you're welcome to join okay

let's so let's talk about the aims and the claims of the paper

um blue do you want to start by saying anything about just your broad overview of what you thought about the aims and claims before we read some of the quotes from them

uh just that i liked the paper i thought it was really um

well written and made the fep very approachable

for a variety of different backgrounds and i can you know read some of these off if you want so the first aim is according to

predictive processing theories the generative model is literally implemented by a human brain to calculate the potential states of the

environment

termed a realist approach whereas other approaches take it to be an insightful statistical description that a non-scientifically trained organism has no access to termed an instrumentalist approach

in this paper we shall engage with both debates and argue that the representationalism debate is not relevant to the fep

and realism is doomed to fail regardless of whether it is representationalist or not

and conversely instrumentalism can drive either way or so we shall argue

thank you for reading it so it's saying this first selection

is kind of laying out the the spectrum or the battlefield

or the continuum of approaches or understandings

whether we take the fep to be more realist or more instrumentalist and we're going to go into a lot more detail about that

and in the middle quote they basically write that this representationalism debate is important

but it's not relevant to this realist instrumentalist contention

which we're going to graphically see soon and then their conclusion

is represented in the third part and their conclusion is that realism is

doomed to fail

independent of the representationalist question and instrumentalism can work

so those are the aims and claims and the way that we're going to be understanding the paper's meaning and whether or not it makes sense or lives up to its claims so to understand whether it lives up to its claims and to understand how those implications would be in our own lives as well as in the broader literature we kind of have to understand where this paper is situated so what is it related to what other areas of research is it uh going to be citing what else is it discussing in common with other areas so we know that this is a free energy principle related paper but also the keywords of the paper were provided as scientific models representationalism realism and instrumentalism so these are some of the main keywords that we're going to be talking about and the goal of these keywords which is just again you could take a course or do so much independent research on any of these topics so we're just going to be broadly going over these a few different keywords just very at the surface layer but picking out a few good examples and that's just to connect the active inference ideas and research community out with other areas that might be useful for applications or for insights and then

also provide
accessible on-ramps into active
inference for those who might be
familiar with
one of these keywords but not with
active inference itself
so first is the scientific models
so scientific models uh blue
how would you describe a scientific
model because i was almost um
just tabula rasa with looking up images
on
image searches because like how does one
go about saying what a scientific model
is or how does it
play out for you so a couple of things
that i always think about when
i think of a scientific model is the
quote from george box that's
that um all models are are wrong some
are useful
uh and then i always go back to the book
that i read that was a compilation
called uh the map is not the territory
and so this really you know plays out in
the way that like
in constructing a model what we are
doing is essentially building a map
whether it's a map of you know gene
states or
or regulatory patterns or neural coding
or whatever it's it's a map and it's not
ever directly the territory because if
the map was the territory it would not
be useful
right so you need to have like so there
has to be some dimension reduction
in in a scientific model but not too

much otherwise it's it's overly
simplistic and
doesn't encompass all of the potential
variables so that's what
modeling like really brings up for me
nice and there's a there's a borges
story
about the map that just pastes over
everything it goes into people's lungs
and it kind of
becomes one-to-one with the real world
and then it you know is it the real
world
so we're dealing with not that situation
but we want to be talking about things
that we
care about and want to explain predict
design control
perturb we want to be useful in the
world so we want to generalize from our
experiences and
look at empirical observations in a way
that's nuanced so that's kind of
scientific modeling
and if you look on search engines or in
the stanford encyclopedia of philosophy
you could get a few different
distinctions like physical models
the kind of thing that a artist might
use as a template
like a torso with some clothing on it
and then mathematical models which are
the kind that we're going to be talking
about today but also there might be
qualitative models or models that don't
use mathematical symbols but
are still conceptual so in the stanford
encyclopedia of philosophy just table of

contents

at the sort of overview layer we can see
that there are several ongoing
related fields of philosophy that are
talking about scientific modeling
ontology what are models epistemology
how do we learn about models what do
they do for us cognitively
and then how do we think about different
kinds of models and

mel andrews paper in active stream 14
discussed a lot of these topics so let's
talk about the scientific
models of a few different types and this
is pretty central
to the discussion of paper today which
is

the uh descriptive of scientific models
so we're talking about
family of models and you have the taller
and the shorter
within the family so now we're going to
talk about the more realism
or more instrumentalism in terms of
scientific models

so um blue do you want to read the
paragraph
sure realism and instrumentalism here
concern the models and statistical
manipulations that make up the fep
and whether they are thought to be used
and manipulated by the systems under
scrutiny independent of scientific
inquiry which is rea
or conversely whether they are thought
to be scientific tools
wrought by humans in specific
socio-cultural environments to study

particular systems which is ins
with rea being realism and ins being
instrumentalism
yes so there's a continuum from extreme
realism
with a rhea acronym to extreme
instrumentalism
and then in the middle you could imagine
is something that draws insights or
tries to be sort of like a compromise
between the positions
but we could ask whether a theory or
whether a model were
more amenable to one of these sides or
another
and the archetypal realist theory is
saying that
the model is what the system is doing so
let's think if you had a pendulum
and your model had a gravity force and a
hinge
and you might say although these are
just parameters with the angle
and the force of gravity and the weight
of the pendulum it actually has a
natural interpretation
i i don't know and i don't want to go
too down the rabbit hole with what's
realist or what's real because
blue you know that it's an open-ended
question but
the instrumentalist one would be more
like i'm modeling the stock market
and so i'm going to model it like a
pendulum with a weight and in that case
you wouldn't be saying
the same kind of claim about the
pendulum so if you have a full model

of what you think is actually happening
out there it's more realistic
and that doesn't mean it's more accurate
per se it doesn't mean that it's cheaper
to run
it's always in balance with a lot of
other features of modeling
but this realist instrumentalist aspect
has big implications for the fep but
we're just bringing it up
in the context of what the idea is by
itself
for now and then we're going to connect
it to the fep with the paper
anything else to add there just like in
the
author's interpretation of realism or or
at least what i interpreted that the
authors
interpreted um you know the realism like
viewpoint or stance is actually
it seems very mechanistic like in
science we're always or especially in
biology
like what is the underlying mechanism
instead of like building a big
model we're like where is the how does
it work and so
this this realism claim is actually you
know
it says that that this is actually the
mechanism the model is
the way that it is working so so you
know i think about gene regulatory
networks and some things that might be
amenable
to that kind of like realistic
representation

um as opposed to instrumental which is
which um you know look like
there's a useful you know like a model
of the stock market or something like
that
yes so let's talk about this other axis
so we had sort of one continuum that
categorizes or describes
kinds of scientific models and then here
is going to be another
axis and we're putting them orthogonal
or just at 90 degree angles to each
other because you could kind of pick and
choose we're going to see how that plays
out
and this is a debate or a continuum
known as representationalism and it's
related to a lot of other topics and in
earlier discussions with
alex kiefer with maxwell with others
we've
broached this idea of what is a
representation is a representation
mean that it has to be inside of the
organism or can it be a distributed
representation um but that's
kind of again what we're locking in on
with this
question is what exactly is a
representation
and does the theory highlight aspects of
the world that are
like representations like a specific
pattern of neural firing or
a specific compound that's that signals
or represents alarm
um or do people take non-representation
stances blue what would you say is

differentiating or
relevant or what's on the table with
this debate
so i i feel like i i am like personally
very
very biased here i think um you know
what's on the table is is there you know
are we perceiving
the object in perception right i mean as
a neuroscience background it's like are
we literally perceiving
this orange which is round flavorful
juicy like cold it might be lots of
different things are we perceiving this
orange or are we perceiving whatever our
mind represents as this orange sweet
juicy like brightly colored flavorful
scented whatever so are we perceiving
some representation
or actually the orange directly so i
think that that's that's the debate
the good news about this debate is that
for this paper it's going to turn out
that
we can be agnostic so if we're not sure
or we don't know or we don't even care
it's going to be okay for this paper
let's look at the abstract of the paper
with a picture by sasha in the
background of neurons
the authors write the free energy
principle provides an increasingly
popular framework to biology and
cognitive science
however it remains disputed whether its
statistical models are scientific tools
to describe non-equilibrium steady-state
systems

which we call the instrumentalist
reading or are literally implemented and
utilized by those systems the realist
reading
we analyze the options critically with
particular attention
to the question of representationalism
okay blue and then on the second part to
that slide of the abstract can you read
it
sure however it remains disputed whether
it's statistical models are scientific
tools to describe
non-equilibrium steady-state systems
which we call the instrumentalist
reading or are literally implemented and
utilized by the systems the realist
reading sorry you just read that
we argue that realism is unwarranted and
conceptually incoherent
conversely instrumentalism is safer
whilst remaining explanatory powerful
moreover we show that the
representationalism debate loses
relevance in an instrumentalist reading
finally these findings could be
generalized for our interpretation of
models in cognitive science more
generally
nice abstract and pretty interesting
topics let's look at the roadmap to
see how do the authors take us from
a to z and then we're gonna go into the
figure
the table and go through a couple of the
key quotations where if you read over it
and you weren't
sure you know what were the key turns

the parts on the actual road trip where
someone says
make sure to take a left at the red barn
or whatever what are the parts where
you want to really evaluate these
sections and ask how you feel about them
so we're looking at the roadmap and
it starts with an introduction before
jumping right into the free energy
principle essentials
and i think it will be fun to go over
the quotation and the figure there
because
blue i know that you wanted to explain
and think about it in a few
new ways and then in section three
there's the discussion of
representationalism so it starts off by
discussing representational realism and
then
talking about non-representational
realism we're gonna look at all this
quadrants all the two by two soon
in table one we see a actual summary of
the possibilities the two by two
possibilities of instrumentalism versus
realism
and representationalist or not and then
this is brought back to generative
models in the free
energy principle and then section 4 is
sort of a plea for instrumentalism
and talks about how independent of
whatever we think about
representationalism
instrumentalism can work for us and then
we're going to talk a little
bit about what that actually looks like

and then there's a conclusion with some really awesome

philosophical and broader points any thoughts or comments on the roadmap

no no let's do free energy principle

essentials so the free energy principle

according to the authors this is the

quote is based on three aspects

blue how about go through the three

aspects just what was interesting what

did you like about this

layout the first aspect is the

observation

of self-organization and you know in all

biological systems

are our non-equilibrium steady-state

systems because if we were

in equilibrium we would all be dead

right so we're maintaining this constant

influx and

and efflux of you know nutrients and

waste and

all of the things that that are

necessary to keep us going um and so

that

is the first uh aspect and then the

second aspect is that living organisms

can be described as stochastic dynamical

systems

possessing attractors and i talked about

uh gaining energy which i thought this

was cool so you have like the the lorenz

attractor here um and in this space like

you know

if you add energy it like brings a

difference

like state that the organism can be in

right so so the addition of energy

you can add start adding points in this space and like in infinite entropy like the whole trajectory would be black right like you could occupy any point in this phase space so so that is you know the expansion of the phase space by the addition of energy and then the you know the reduction of that um space is an increase in in certainty right like you can only you know once you start down a developmental trajectory like there's only so many you know like paths you can follow and so it's your you know trajectory has been reduced contracted your space has been reduced in that in that way and then the third essential point was that the states in which the self-organizing system is is at a point in or are in at a point in time can be identified by the interactive role that they play within the multi-level self-organization scheme and like you know daniel that i'm very uh interested in the multi-scale uh possibility of agency and and systemness and and all that so at every level you know from the molecular level to the cellular level to the brain to the organism uh to groups of

organisms and so forth i think that
that's all
you know those are all self-organizing
systems
nice now where does the free energy
principle come into play
or where do we go because those are
three of the coolest
aspects of complex systems it's probably
what drew us both to
complexity where we met is through being
attracted to these kinds of questions
because it's kind of like a community of
people who
like topics like self-organization
dynamical systems and chaotic systems
strange attractors
multi-scale systems emergence like those
are kind of the key
pieces so i like that the fvp
starts with it's it's being based upon
these
observations and then it's going to
build on
that so it's kind of like taking the
complexity
not as an endpoint but as a beginning
point so that's one thing that i think
is really fun about
fep so how are we gonna bring fep into
this
specific setup as far as here's what we
know that we have to explain
we don't have to explain a single or
double pendulum we have to explain this
so where does figure one come into play
in relationship to those aspects or
anything else

well so let me add to this slide first
so the the fep where it really comes
into play is is because it's focused on
entropy reduction right it's focusing on
you know that this the state that the
organism assumes
is is that of of the reduced entropy of
the least entropy right so it's the the
minimization of free energy through
entropy reduction
so so that's that's that on this and
then on the next slide
um you know as we get into figure one
they they talk about the um active
states and the sensory states
of organisms so this really will go into
the free energy equation although the
equation's not discussed here they do
discuss some elements of it
so there's internal states of the
organism active states
sensory states and external states of
the organism and so this is like really
the patterned activity of every organism
it's perceived through the environment
uh the the active states are really what
the organism is doing
and then the sensory states are what the
organism is perceiving
and then there's blanket states right so
the blanket states are
all the hidden states of the world the
active states
the noise of the random fluctuations and
this is
the internal states and the blanket
states go to minimize the free energy
those two things together

so this is like you know how it's a
formal way the free energy is a formal
way of
measuring the surprise and surprise not
in the information theory
uh way not surprised as in like an
entropy reduction so it's important i
think to make that distinction
um but but it minimizes surprise between
the prediction and the the actual
reality
so the surprises is like that divergence
right the
kubrick kublai blur divergence between
those two um you know the surprise
uh is is that the divergence between um
the posterior and the essentially
thank you thank you for that let me try
from a different angle yeah
sure so yeah we're coming from the
scientific modeling literature
not the graphical bayesian literature
those are cited
but in this paper it's being approached
and kind of
reflected upon from this philosophy of
science and almost history of ideas
away so last time in 14 we talked about
wanting to cut nature at the joints
because we wanted to understand what
kind of things were what kind of things
things were and we wanted to be able to
model them effectively
now we're going to be thinking about
using this
markov blanket formalism that partitions
external and internal states but that's
kind of really like saying that we're

defining an interface and there's two sides to the interface because it can be symmetrical in a sense so we're defining an interface which implies an internal and external partitioning and this is going to allow us to do modeling in a way that is multi-scale because different kinds of nodes internal can be internal to each other so you can have hierarchy or nestedness of modeling so it's kind of like like having a ton of small little computers running and then another computer could be like on top of them coordinating them so those kinds of hierarchical models can be done with this type of uh markov blanket interface definition framework because you can have internal states that consist of multiple things so that's nesting hierarchy you can also have hierarchy or collective behavior or interactions between subunits because the external states of agents or nodes can be other agents like themselves or different things like the environment so there's this massive amount of systems such as interacting agents or agent environment feedbacks that you can think of with an interface in mind and then it lets you

get at this question of modeling in a way that's multi-scale can deal with arbitrary dynamics you could have the internal states could be arbitrary it could be a flat wave it could be a sine wave it could be a jimi hendrix song you know it could be anything because it's defined in kind of the most skeletal way so this is like the body plan for one layer but then in other papers by maxwell and ramstad and others we've seen how that kind of plays out in a multi-scale way because that's like the beyond internalism and externalism paper where we talked about this in the multi-scale way and then that was just on one slide ago on the bottom right was the multi-scale question so we know that we're dealing with multi-scale so when you're looking at figure one it's kind of like you're thinking about one ant interacting with the environment but just know that we're fully taking into account this multi-scale approach that's also gonna come along with the generative model and this action selection the policy and the surprise so there's a lot of features to it and this is a simple figure we've seen these kinds of figures before and there's always more things to say about it and also if people in the live

chat have any
thoughts or questions we can say those
or anything
anything else to say on one blue no i'm
good
nice it's just it speaks a lot that
with one figure two very thoughtful
authors
put this because there's a i don't know
if it's
uh citable or not but you know data on
how even
one equation will reduce the amount of
people who will like read a paper
sad but true so when putting figures
with math
it's just it really has to be done
appropriately and this is an example of
that's very elegant in what it conveys
and in it what it opens the door to
so whatever you're seeing in the figure
it's what's there
the caption just describes what the
parameters are but again
whether real systems are doing that
realism or whether this is just a
graphical model
that we're applying like any other
graphical model
in statisticalandia that's the topic
of this paper so if you want to learn
more about the technical details then
that's where the citations lead you
but we're looking at this and we're
asking what would it mean if the world
was actually doing this what would it
mean if this was how we could approach
systems like a scientific instrument

that's the discussion blue so so i do
think that this
this figure like you know it just takes
all of
everything that you've ever known about
the free energy principle and like puts
it in front of you in a very simple and
elegant way like it's like
you just want to unload like brain dump
all of your your
fep knowledge at a simple figure and so
it's very you know minimalist which lets
you
it makes it accessible right for for
people whatever their level
of you know mathematical background
which i think is really cool
if i were going to read this by the
numbers by the colors
so each of the things has a slightly
different color
so that's reflecting that they're
similar like they're squoval
boxes but they're different types of
things which is true
um they have a single letter in the
equation so each of them is defined by a
single letter
that's related to some other function so
it's kind of like you know
x number of pizzas that we have to order
equals dot
it's like one of those things but it's
related to that color of the squoval
and then there's the same kind of
relationship with a barbell
that relates internal and active states
as sensory in external states

and then there's a similar kind of relationship how sensory states influence internal states and active states influence external states and then there's a sort of relationship between with an arrow a new kind of relationship between active and sensory states so this by headed arrow to me in the middle is the definition of the interface because as we talked about in 14 the markov blanket had the interpretation of just statistically insulating nodes and then later on fristen separated that into incoming or sensory states outgoing or active states and then internal states act uh impinge upon the interfaces active states like that sort of like you know the internal we can think of as the brain and active states being the arm the same way the brain sends signals to the arm that's the way that the ecosystem the external states is like sending signals to the sensory modality so that's the trees green going into your retina just at a first pass so the internal state influences action states just like external states influence sensory states what do sensory states do they update internal states and what do action states do they update external states so there's so much that you can really say and i agree it's a minimal prompt that helps

coalesce a lot of understanding
and there's just four boxes and a couple
of lines
okay table one so this is the kind of
the crux of the debate and it's awesome
to see a philosophy paper with a table
because it lays it out starkly so
instead of just saying
this chapter or this subheader or this
paragraph
this is a and b this is a but not b so
here we're just seeing it in a graphical
way and with acronyms and the letters
kind of look similar a little bit at
least to me so we're gonna
colorize them um and also just make it
very clear which one we're
talking about but basically it's a two
by two so kind of like a punnett square
and or an eisenhower matrix and it has
the two
options for each of the dimensions that
we care about so on the top
columns are whether we're talking about
realistic realism
theories models that are have realism or
an instrumentalism
approach to modeling in the fep and then
the rows
are whether we think about things
representationalism
so do we think there's representations
occurring or non-representationist
and then you have the four combinations
of the three letter schemes rep
rhea et cetera et cetera so what would
you say about that blue before we look
at it a little more

uh visually uh just that
it's very um cool how they kind of
restrict the representationalism and the
non-representationalism and
because they're gonna really take that
out of the
equation at a later time so um not
really much else to say
but other than you know that there's
different combinations that can exist
right the representational realism
rep rea non-representational realism
which is nrp
rea representationalist instrumentalism
and non-representationalist
instrumentalism so the representational
standpoint representing
like where you're using a representation
of the object as opposed to the object
itself like where you're perceiving
the orange like your mind map or your
mental map of the orange versus the
actual orange
uh and then realism and instrumentalism
being um
you know the realism is implemented
directly
and instrumentalism is implemented as a
statistical tool or model
something like that and if it's too many
syllables
just think realism yes no
representation yes no or there's longer
words on the other
sides let's look at it in a graphical
way
and also it's it's it's not um for
granted

that you could make a two by two or make
a orthogonal axis
because there might be two philosophical
ideas that are correlated like one
implies the other
but for the purposes of a paper
sometimes what somebody will do will be
lay out two options like um the plant
could either be influenced by nitrogen
or not
by light or not and then you use that to
study what the plant needs or how two
factors are related to each other
well here we're studying how two ideas
two continuums of ideas
are related to each other and so it
might turn out that some quadrants or
some combinations
are mandated some are prohibited some
are
not preferable or preferable for some
other reason so you lay out the two by
two
or you lay out the higher dimensions of
combining thought
and then you ask what are the pros and
cons of each of these areas
so here we have the x-axis is the
realist
on the left instrumentalist on the right
and then
the fep options for representationalism
are the y-axis
representationalist higher and
non-representation is lower
and drumroll the contention
of the authors as screenshotted in this
section 5 conclusion so always awesome

when the first sentence of like a
section
actually is a summary in this paper we
have defended the instrumentalist take
on the fep
arguing that the realist approach is a
non-starter so it doesn't go
regardless of whether it is
representationalist or not
so these big x's with a question mark
because we're not saying it's the end
that's not our take it's not even the
end point for the authors
or the community per se just that's what
the paper is about
is saying no matter what you think about
representationalism
so no matter where you fall on the
y-axis
realism is a non-starter so realism is
not going to work
anything with rea is not going to work
anything to the left on over this is not
going to be a tenable
approach for the fep and we would do
better off
to hang out on the right side here with
instrumentalism where there is a debate
or there's more to be understood about
representationalism
but let's be clear that fep is being
approached
instrumentally rather than with a
realistic research agenda
or interpretation paradigm what does
that make you think about blue
just um you know in some ways like
like it just it brings up a lot of stuff

right everyone has like maybe what they think

so in the instrumentalist um

you know from that framework right it's

it's a model or a tool that we can use

not something that's actually

implemented like say by the brain right

so so the brain's not actually

using the fep to do thought like

processes

but you know you know we can always make

it real right like so

so we can take the you know statistical

tool or model and put it into a computer

and then it actually does use that right

and so so then

you know there's always ways to you know

abstract realism from

from the instrumentalism uh you know in

constructing something that runs on the

model

maybe that's not biological very

interesting question so it's almost like

if you had a state machine

um and then you had a mathematical model

of it it would be

almost a realistic mathematical model or

it might be a comprehensive

and then yeah when we um and we talked

about the metaphor

reification and about how sometimes when

there's a conflation of the map in the

territory or there's only one map for a

territory

so people start navigating off of that

what does that do

and then how do we know that we're in

that situation that we're in the

epicycle

situation where there could be a
breakout into a whole different way
of thinking on a topic but we're just
seeing within the
bounds of how we think about the system
really being or what instruments we
really have access to

let's look at some of the quotes because
it's worth

going into if the fep is an
instrument

shop if it's more like a music shop you
know than the

the hall of reality what are those
instruments

so what will the fep instrumentalize as
how's it gonna gonna be manifested and
then what will allow us to do

why does that matter because if we were
going down the realist rabbit hole

we could ask questions like well what
does that mean for the meaning of life
or

what does that say about how we should

live a good life or how does that relate
to the trolley car problem or something

now we could still ask those questions

but if we first go the instrumentalist
route we'd be like saying

okay let's model the trolley car problem

not thinking that it's actually this way

but

and then you could still be clear about

where your non-fep derived

value system arose from because just

like any other model linear aggression

anova it's incomplete without a little

bit more

on that side and that's really something
that's coming more and more into focus
with ai ethics and

how data is being utilized in society so
it's pretty interesting to talk about
those things all right do you want to
read this

full page quotation with the aim where
the authors again thankfully lay out the
aim of their paper

sure the aim of this paper is to show
that philosophically

instrumentalist thinking is less
controversial yet remains explanatory
powerful and can yield important
insights in an organism

environment dynamics an instrumentalist
attitude about the fep is a safer bet
without losing the potentially high
returns

after briefly describing the fep in
section 2 we assess two proposals
made in the realest logical space that
of representational realism

and non-representational realism in
section 3. we reject both of them and in
section 4 we proceed to offer positive
reasons to embrace instrumentalism about
the fep

given the activity dependence feature of
neuronal activity

dynamic causal modeling under the fep
seems to be the most suitable and
promising set of instruments

to preserve the character of neuronal
activation as we empirically know it to
be

activity and coupled systems from this
angle

realist arguments look like forcing the
world to conform with the
anthropomorphic instrumental lens we use
to make sense of it

cool so one note is the defining two
kinds of realism

representational or not again we're kind
of rehashing some of the same
discussions because the strange
attractor is the paper we're gonna keep
returning to this because you could
study

your whole life so if you think that you
haven't heard the word

enough in two hours then wait till it's
been thinking about it for two years
and this reminds me of just going to a
restaurant it's like you could just say
no i don't want

soup and then here it's like there's two
kinds of soup and i don't want

soup a or soup b it's like you divide
to conquer instead of just saying we
think instrumentalism is better than
realism

paper's done they actually said there's
there's more there's nuance to the
realist position

and we're going to give a strong
presentation

of why somebody would honestly believe
that realism works for various reasons
and then we're going to individually
respond to the

uh unique realist perspectives maybe
there's more than two

there's other axes that you could go into uh defining but we're going to actually uniquely respond to why different kinds of realists are realists rather than just only looking at the realist instrumentalist number line and just saying instrumentalism's better because it's more useful duh that's less nuanced paper and it's less useful because it doesn't really help us understand why realism is being dismissed unilaterally out of hand this is a much more nuanced way to deal with the question and then also i've highlighted the dynamic causal modeling because it's going to be discussed as kind of the leading instrument so again if the fep is to be understood in an instrumentalist way so it's the music shop with the instruments what are the instruments what have people done with them do they not exist is it an empty music shop where people are saying just come and build music things we don't know what it's going to look like yet or is there something already there and could there be something that looks different or is it going to only look similar those are the kinds of questions that in early 2021 were wondering aren't we any other questions or thoughts on this blue

before we go into
dynamic causal modeling a little bit
more in detail no let's take it away
yep okay so dynamic causal modeling
we often hear things like well where's
like the textbook on
fap or active inference or what are the
sections that are related to understand
it
or it's something that we all want to
improve the state of and
again it's one of the main projects in
our lab to be developing educational
material and
bodies of knowledge that being said if
dynamic causal modeling
as posited by the authors really is the
leading
implementation of the instrumentalist
fep
then it would make sense to understand a
little bit about
what dynamic causal models are because
these are ultimately the form
or the type of models that you're going
to be discussing
so if we've dispensed with the fep as
describing how the system
is with a realist lens we're going to be
modeling the stock market using active
inference
using the fep but we're going to be
using it instrumentally we weren't
saying that the stock market
was linear aggressions or was arima
models so we're not saying that it is
fep we're just saying that we're going
to explain variants

we're going to explain design predict
control better
because the fep so what will it look
like beyond the metaphor
and just thinking about it in a cool way
well
it's going to look like dynamic causal
modeling that's what is going to be
discussed in this paper so this book
from 2007
i think it might be the most recent
version statistical parametric mapping
or spm
is a long running tool kit that's been
developed by
carl fristen and others in relationship
to
this dynamic causal modeling framework
so dynamic causal models are kind of the
statistical family of models
and statistical parametric mapping is
first off spm that's the name of the
toolkit in matlab
as we've discussed in the model streams
python and other
languages are being developed for this
as well so we're not going to go into
that right now
it's called the statistical parametric
map because
given the dynamic causal model that's
being inferred
this spm package puts a layer of
statistics over that
and it asks from a parametric statistic
perspective like a p-value perspective
just again this is a whole like 800 page
textbook so there's just like

two sentences it does statistics on
these kinds of models
so that entails not just specifying
these models but specifying
what fields of models they exist within
and a lot of other attributes it's not
the same just to write out one models to
do statistics but that's what this book
is for

now what is the form of these models
there's other

cousins and colleagues but this is one
of the core forms in dcm

there's two coupled equations the first
top equation

has z which is neural activity we can
see from this caption

that the change in neural activity so it
could be dz or $\dot{z}$

just its neural activity is hidden it

cannot be directly observed using
non-invasive functioning

imaging modalities so it could represent
neural firing but it can actually

represent just

any hidden underlying unobserved state
and that underlying hidden state is a
function of

itself at the prior moment and in the
past that's

the previous z its own value basically

then you are the experimental inputs

so that's the things you added nitrogen

and you're inferring the ecosystem

so that's this kind of idea and then

θ is the number of parameters

in this fitted model and again we're not
talking about reality

we're instrumentalist so these
parameters are just a heuristic
8 is better fitting than 12 okay then
that's how many we're going to use we're
not making claims
we're getting hung up on that this z
hidden underlying equation that has to
be fit
just like you didn't observe the
regression line you're not going to
observe the dynamical z
but you're fitting it with data then
what data are you fitting it with
you're fitting it with y which is
the observed activity it can be unimodal
or it could be multiple kinds of
time series but it can handle multiple
different types of time series together
like
eeg fmri meg with different types of
time
series being overlaid with different
types of uh dynamics and sampling rates
and everything
and that under that observable y
function is a different functions g
related to inferring this underlying
hidden neural state
plus parameters of the statistical model
with a noise term
that can be modality specific so
that's almost like if it's really true
that the fvp is instrumentalist
then it's really true that we should
think about things this way
with like a hidden component here with
change in hidden variable
being related to the hidden variable

state and other features inputs and
other features like attributes
and then we're observing and it turns
out that this
minimal two equation set is going to be
really helpful because it allows us to
make an arbitrarily complex
z whether we know nothing or a ton about

z

we can arbitrarily lay that out and then
we can have y
be very scarce or very noisy or
just only have y for 15 minutes so it's
going to open up

a lot of types of models that are
with one aspect that we didn't see and
then a semi-related aspect that we did
see

okay any thoughts on that blue before we
look at a few dcm examples
no go ahead let's go okay so this is
an example of going even more one layer
more specific now we're looking at
a dcm paper looking at specific brain
regions

so one thing that's really important
about the dcm

framework is where it had the
experimental inputs so again we're
instrumentalists not realist so we don't
think that these connections
reflect like where the neurons are
touching it doesn't say this is what
causes what

it's parameters that we're fitting in a
model and so

the way that you ask does a influence be
in the dcm framework is is it a better

model

when i have a statistical influence

between a and b

rather than the statistical model where

that's absent which by definition is

simpler

but so if it explains the same amount of

variance then the simpler model is

preferred

however if there's a significantly more

amount of variance explained by the more

nuanced model

then it's equivalent to saying there is

statistical evidence that a

and b are influencing each other and so

this is

something that's really cool in this

textbook and in this paper

where they talk about psychophysical

interactions and so

in this case they're looking at a few

different brain regions

which are the nodes like stg you know

superior temporal gyrus and a couple of

other

visual areas and visual associated areas

and then each of those underlying hidden

actual

neural regions are emitting a y which in

this case is the voxel

of data for fmri so there's like a time

series of intensity with a bold signal

we're not going into fmri but it's a

time series of intensity

that's emitted by an underlying

something

that could have a very very different

profile than the actual emitted signal

because of how fmri is measured and all these things like that then they can ask in two conditions where the same visual input is happening so it will be like a field of things moving or there's different visual assays again we're also not going into that you can have two conditions one of them where the participant is told look at the center and the other one where the participant is told do this something else you know look at the far side or track the moving dot that is going to be reflected by changes in um the visual region potentially as well as in other regions and then they can look at two different instruction sets to highlight whether there's a statistical association between two other areas of the brain again it doesn't say anything about whether they're actually connected with neurons but it highlights a dynamical or a functional dependency doing these kinds of nuanced statistical models so it's kind of like a structural equation model but with time series and dynamical series and partially observable states and uh bayesian networks and a few other things next slide blue or any other thoughts no i'll have a thought at the end but then okay there's two more i think dcm slides this is just to we're not going to go through

this one

this is just like saying that was the

simple that was 2003 dcm

with the uh the v1 region this is 2019

dcm dynamic causal modeling revisited

narrow image first in

all 2019 so it has become implemented

with

things that are very interesting

and it's a very nuanced model so look

at what the developmental trajectory has

been and it kind of makes sense

how more and more features like the

presynaptic and the postsynaptic

dynamics

could be incorporated within this course

scaffold so that's why it's really

important to start with the right

generalizations because then it can get

nuanced

you know 20 years down the line and then

this paper just the last point on the

dcms just to give one example of the

applications of dcm

because again if it's instrumentalist

then

we want the instrumentalist uh uh we

want the instruments that are useful

we want the financial instruments

musical instruments carpentry

instruments

that work so what can we actually do

with dcm

and papers like this one in 2013 and

other more recent ones

show how we can actually use free energy

fitting of models

so we talk about free energy

minimization on big data sets

or we're talking about whether there's
free energy minimization happening in
the context of organismal behavior
again instrumentally not realistically
that

free energy minimization can be
understood as navigating this trade-off
with accuracy minus complexity and so
we're going to use this free energy
minimization

on graphs of dynamically associating
brain regions

as acid by neuroimaging and then use
free energy

to fit this vast family of models down
into something that's on that manifold
that's kind of like pareto optimal
with respect to accuracy minus
complexity given the experimental design
and the massive amount that we actually
do know about things like hemodynamics
and the bold responses error dynamics so
i'm going to just

stop there blue what do you think about
that

so i just wanted to say like when you
know we're talking about fmri

i don't know if everyone has seen
pictures of fmri and what that
looks like but it's you know essentially
like a black and white picture of the
brain with little areas of the brain
that are lit up

and in in those fmri pictures when you
take that into this realism

versus instrumentalism kind of framework
it was i was thinking of this

specifically as i was reading this paper
it really literally says that that the
fmri
which is like the model of of the
how the brain perceives the world right
so in this fmri scan somehow like you
know the jennifer aniston
neuron like it should be like a tiny
little picture of
of jennifer aniston i just like kept
thinking about that like inside
the brain in the fmri scan we should
literally be able to see
this this realist representationalist
um like the representation of the tree
we see outside should be there in the
fmri scan if it's really
like falls into that that framework and
i just i kept having that thought and it
was like
recurring and silly nice the jennifer
aniston
neural distributed activation pattern
that was the longer article the editor
scrapped the headline
so somebody in the youtube chat wrote
this and it's just great to hear another
perspective on free energy while we're
here
they wrote hi i'm a student in math free
energy to me is just a moment
generating function for a probability
distribution
in feynman's path integration it is a
generating functional
in geometry free energy is a generating
function
for yet more probabilistic information

does this free energy
principle relate to specific
probabilistic information
so let me while blue you can prepare
anything if you want but let me just
work backwards through a few of these
terms just really quickly yes i would
say it relates to probabilistic
information because it's very closely
related to
information theory like we're seeing
gnats on
this y-axis so we're talking about model
fitting reduction of uncertainty
model fitting of probabilistic and
stochastic variables
instrumentally so we're in that
territory and then um
there's i know a lot of detail that
others with a lot more mathematical
background could add to
how the trajectory the path integration
and how the free energy is like a path
integral
that's being optimized as far as policy
because you're not just doing
the instantaneous static optimization so
it's not like just that first moment
it's
more like a trajectory of a dynamical
system and then you mentioned it's a
generating function
for more probabilistic information
there's probably multiple ways to take
that but
it seems like we're talking about
generative models and the way that
generative models even in feedback with

their environment
could be generating and uh more
information and
doing behavior what do you think blue
um i think that you summarized it quite
well and i also
i do think that um it's important to
you know distinguish the surprise as the
authors did in this paper that surprise
in the free energy principle is not
the surprise from information theory
just make that one
kind of distinction because it's easy to
conflate those those two things yep and
as far as feynman specifically i know
that's the variational
principle and so we're heavily in the
variational territory
and variational base vb but
it would be awesome to have somebody
with a little more background to answer
it more
speaking of bays and of models and also
of everyone learning and continuing to
be on their
active inference journey how are we all
going to be
challenged in learning because there's
so much to learn in philosophy and in
math well
it's awesome when the paper can give us
both philosophy and math together
so even though this is a simple equation
for some
for the two equations that are in this
paper we're just gonna look at them and
understand
what they're actually doing in the

context of the paper

because again they're speaking to an audience including philosophers and they've gone out of the way to include equations

because this is the clearest way to signal what they're thinking about so here they're talking about rep rhea so representations and realism so yes for both and

um they write that this rep review comes in the form of

process theory offshoots of the fep such as predictive coding predictive processing

or more generally pem theories of cognition

what is pm prediction error minimization right

okay by employing bayesian epistemology or using bayesian ways of looking at things

scientists refer to the model of the nervous system by using technical terms pertaining to the basis theorem as shown here

so blue what would you say about bayes theorem

here so uh just that

you know to your point about philosophy and math i think that that

really you know this equation um

even for someone like you know i i'm teaching my nine-year-old daughter how to do math and she would be like that's not math

so really like there's they both come down philosophy and math to

to formal logic which is um really
kind of highlighted in um in bayes
theorem so
the theorem really to read it it's
the probability probability of a given b
which is
the left side of the equation is equal
to the probability of b
given a like you know if this then that
right the probability
of um you know one event in occurrence
with another
uh times the probability of a divided by
the probability of of b
so so that is the the um the reading of
the equation and it is like formal logic
like
you know what are the chances of trying
to think of a good
example um if
uh you know the eagles win the super
bowl
then their quarterback goes to the pro
bowl like what
right like so what are the chances of
what is the probability of that
quarterback going to the pro bowl
giving given that team winning the super
bowl like what are the chances of this
given that
and what are the you know and it's the
probability of what's the probability of
the eagles winning super bowl
divided by the probability of um the
quarterback going to the pro bowl so
those kinds of of things
like if one event then another here's
another take on that is

when events are independent we know that they shouldn't impinge on each other like flip a coin you get heads you roll the dice you get a six so it's like one half chance of getting a heads and then one and six of getting a six so it's like a twelfth you just multiply them together so that's regular probability numbers between zero and one nothing happening or unlikely to happen and extremely likely or must happen with one so if they're independent you just multiply them but this is actually for cases where it's not quite so clear that there's independence so it's more for the case in real life where things are associated even if for spurious reasons that's why the instrumentalist reading is so important because just finding that there's some um correlation in the real world doesn't entail any specific underlying model it's just a statistical association between different variables and so here we're studying the set of all interacting factors there's two here but there can be way more than two in large bayesian models for just things that influence each other so we're thinking from the simplest case with just

two non-interacting events and then you could think about more than two non-interacting events and then you might want to ask how events interacting makes it more complicated and of course it does make it more complicated but that's why there are good instruments to deal with it like spm and the dynamical equation modeling approaches so that's bayesian approach and you could learn a lot and there's courses and lectures and stuff on bayesian statistics but this is kind of the core of it and the point is that the terms and the uh frameworks that are used of in the rap ria camp involve this equation in spirit and in code so they use terms like the prior and updating the prior and calculating the posterior which is a few sections of these equations which we're not going to worry about but also they implement scaled versions of these equations on empirical data let's go to the second equation so the the previous equation was about updating with new information based upon priors just taking what you knew about how the world was associated implementing information that's gained from experiments or observations and updating your model of the world in some way now we're going to be talking about the

kind of different area of math
a different feature that equations are
helping us understand
so do you want to read the equations or
anything else you want to say on it
go ahead uh yeah i'll read the the quote
if you want and then you can explain
um i do want to say that the probability
of the eagles winning the super bowl
this year is
zero so just by the way
notably notably noise can drastically
modify
the even deterministic dynamics
importantly
this means that stochastic dynamical
systems accounting for noise are
equipped at least in principle
to capture how existing states
contribute to adaptation
state space models are among the most
suitable set of techniques
uh from rozzly and first in 2016. to
model the unfolding activity or behavior
of a system subject to fluctuations in
noise
described by an ordinary differential
equation
and then i'll let you take it from here
down sure so in the previous equation we
were thinking about
systems that change through time when
new information comes into play
now we're thinking about other aspects
of systems that we care about
specifically fluctuations and noise
so that's kind of like change through
time so it's another way to say that

it's changing
and developing through time in a way
that either oscillates very cleanly
or it could just be all over the place
it could be any kind of rough path or
any kind of signal anything in the
signal processing
domain is going to be considered a
fluctuation it's just not
a flat line or that could be considered
a very small fluctuation
so things that are changing through time
and noise
so we're kind of getting the basics of
statistics things that influence each
other
things that change through time and
things that have noise
which is to say that in the measurement
or any other
um step between the system occurring
and you modeling it which is to say
everything there's some
lossiness and therefore there's some
noise term that has to be modeled
so again the instrumentalists take isn't
that the noise is like
in the world doing something the noise
term is a parameter
in our instrument that's defined in
relationship to the variance left
unexplained
by the rest of our model so depending on
what you do include in your model and
how you calculate it what data sets you
use
your noise estimates can be different
just like depending on what variables

you include in a regression
which interactions you allow for you're
going to even estimate the sign
of different associations to be
different so if one country has younger
people another country has older people
and then you do a regression you might
think that age was associated when
actually it was country or vice versa
so that's why it's really important to
take an approach that looks at the whole
face space
to understand how to fit these dynamical
models
so instead of descriptive we're looking
at underlying
instead of static systems we're looking
at changing systems instead of perfect
measurement we're allowing for noise
measure uh noise to be included so
there's a few dimensions
where it's like we're thinking
mathematically but we're also
relaxing it in key aspects that help us
deal
with philosophically and just
practically important aspects of the
world
so the kinds of things we want to use
the fep instrument on because if someone
says i made an instrument but like no
you can't play it
because it's like very large or it takes
25 people to play or something like that
we don't want to make that kind of an
instrument so what kind of instrument do
we have
it draws upon these types of mathematics

okay all right let's look at a few more quotations um do you want to read this quote on yet from the fact sure yet from the fact that it is possible to model the process it does not necessarily follow that the target phenomenon represents the intellectual tools we use to model it consider a moving object that can be explained by newton's laws of motion that we can model the movement by that form of formalism does not follow that object represents the law by which it falls few people would claim that the object represents or embodies instantiates implements employees leverages the laws by which it moves because science does not back this up those who wish to do so are committed to a philosophical assumption that moving objects like cells or organs in the nervous system represent laws principles or the intellectual tools we use to describe processes conforming to laws or principles posterior posteriors likelihoods and priors first in whis and hobson 2020 are on the guard on this matter pointing that from the fact that it is possible to map states does not mean that the resulting descriptions refer to entities that actually exist uh and just i love this like newton's law of motion that like i loved this except from the paper

like i was thinking about like you know
the apple classic hole that falls on
newton's head like
does that apple actually like embody or
represent
the laws of motion like does that apple
represent gravity in and of itself
i thought that was a really nice uh way
to put it and it also highlights
why the examples are important of the
example theory
of newton's law of motion and also the
example within that
of the apple falling because if you go
up to someone on the street and you say
you know are scientific models realistic
something you explained them what
realism was and said yes or no
some people might say yeah i mean it's
the best shot or it really is that way
or that's really how the clouds form or
that's really how the climate changes
but then when you say right but then the
apple isn't gravity it's not the idea of
gravity it's an apple
so then doesn't that make you rethink
where you are on the realist spectrum so
this is part of the author's
argument from a philosophical
perspective against
realism they're saying the commitments
that you make
by taking the realist stance on newton's
law of motion
are kind of crazy it's like not good
stances
that you're taking on this example agree
or disagree if you agree

you continue forward and then saying
well then that is a commitment
to also these other implications that
maybe are you still on board but
somebody might say yeah i'm still on
board the cell is just the intellectual
idea of a cell
it's really nothing more than that and
then somebody would say right but then
at some point you're going to be
instrumentally wanting to do something
pragmatic and then you're gonna
like you know have your front of house
be realism but then actually be acting
like an instrumentalist
because in the end you care about useful
models too so this is a long-running
debate but it's just
really captured here let's go to the
conclusion
and then we have a few more things that
we'll talk about so um do you want to
read it or i can't
you can go ahead all right they let me
just
they write in this paper we have
defended the instruments list take on
the free energy principle
arguing that the realist approach is a
non-starter regardless of whether it is
representationalists or not
crucially the question as to whether
systems do or do not model their
environment
will not be decided by neuroimaging
studies or the models we employ
in interpreting the data so this is
awesome it's gonna entail

data and research and empirical studies
but there is no
neuroimaging data set that's gonna
resolve it this paper and following
discussions
are how we get at this issue there isn't
a data set that exists
there isn't a data set that could exist
it's actually a philosophical question
so this is why we want to have
participation from different
perspectives
because it's not a machine learning
question whether the fep
is realist or not and it does have
implications so that's really
a great conclusion this is a
philosophical matter that should be
dealt with by way of philosophical
argumentation
we have argued that the
representationalist realist rep ria
position does not hold up
because of the as of yet missing
naturalistic grounding of
representations independent of
sociocultural practices including
structural representations
citations the non-representationalist
realist position
nrp rhea reports to solve the issues of
rapporia
by removing representational content
from the story yet it does not hold up
because without content there is no
model and no bayesian inference
the instrumentalist does not face the
same problems as they do not ascribe the

modeling activity

to the organism under scrutiny so that's

why they first went to the rep rhea

position

and introduced the bayes equation

because it's really important to

understand reparia

this is sort of like the first take you

say yeah the fep is literally

representing what the organism literally

does it's a markov blanket and it's

dissipating a gradient what else is

there to say

that is critiqued for reasons of

representationalism so then people moved

from the reptria

to the nrp rhea non-representational

realism was like okay well yeah fep is

still

really what the organism's doing

literally realistically

however it's not representing

it's just a representationalist

realistic

model and then according to the authors

take it or leave it or discuss it with

them

later they're saying that that rep rhea

is a contentless model because it's

basically saying you can't have

this kind of a combination of a

contentless model so that's the approach

the philosophical argumentation

approaches they divide into the quadrant

say you know it could be anywhere we

don't really know start with one

quadrant say that it's not good

move into another quadrant say it's also

not good and then in doing so
realize that they've eradicated both
interpretations of realism
on the representationalism spectrum
maybe there's another spectrum that you
want to bring into the picture
but that's the way that you actually go
about asking whether another idea
bears upon this the way that they've
laid it out
so such a great approach and really
clear what would you add to the
conclusion blue
um i didn't really have anything to add
except it was really a good
paper and i really liked the examples
and
the way that they laid it out it's kind
of like you can't um
you know they presented the argument in
such a way that you can't really argue
with that
so i liked that or if you wanted to
argue you'd know where to go
let's just go through a couple of these
questions that you
raised let's just we're not going to
address them but like
what is a cool example of each of these
points or questions or what do you think
is exciting
what kind of project does it bear upon
that these are the questions that we're
asking in relationship to this paper so
first with the process theories
and the fep what was interesting to you
there so this
i i just highlighted this because this

is something that the authors highlighted
um and not really with a question but um you know they they highlighted that the free energy principle uh is um a principle right it's not a process theory and this kind of we discussed this really in mel's paper in 14.0 right so um but but that these process theories such as bayesian brain hypothesis predictive coding active inference really explain how the fep is is actualized right um where the fep you know targets reduction of entropy of the bayesian brain hypothesis and predictive coding focus on maximizing the hypothesis likelihood given the sensory input um and that the fep really doesn't uh imply or require any specific representational tools that are implemented by these process theories and really going back to 14.0 and one which i attended but um going back to that it's you know mel really described it in her paper how the fep fits into scientific theory and um and you know active inference and such yep within the mel so demarcation boundary we're saying yes fep is within science but what is it if not one of these

process theories and it's almost like
fep is the music shop and then the
process theories are like the actual
instruments
that are being played and so maybe one
of them is useful for
one setting or it's useful at a certain
range of volumes or it's useful in a
certain ensemble size
those are the instruments so that's the
instrumentalist
understanding of the fep approach which
is why we can say that
fep is doing things that's
differentiated from other ideas
like it's a different focus on the
reduction of entropy rather than the
maximization of likelihood or
other attributes but what it's doing
it's like a workshop that's a generator
in some ways and we're also constructing
the workshop along the way
but what it's making actually are
instruments and by clarifying that
there's a workshop or an approach
and the instruments that are being
utilized instrumentally
and then making it clear that it's a no
realism
zone we're actually sidestepping a lot
of the questions and moving straight to
the utility
and the ability to translate and apply
instead of waiting
for something that can't be resolved
empirically as we're
seeing from this paper so that's pretty
cool and the process theories in the fpp

has come up in a few previous ones
let's go to ideas and questions that you
had ever thought about related to
reparia
and the prediction air minimization
theories of cognition
so just some um you know of the
questions that were raised by the author
but i think that some things that are
interesting to think about like
does the brain actually perform
probabilistic inference
and does it represent probability
distributions in this variational base
kind of way uh and really what is what
is the agent here or like the the huge
the unit of agency
is it like the brain or the organism or
you know going
to the multi-scale is it subcellular
like the neuron
or the cell or the molecule or also like
groups of systems
ant colonies and so forth like where is
where is the agent
there um and our agents prediction
machines
right i mean really like this is the the
reprieve kind of
take on this but i think it's it's uh
interesting to think about like
does the brain really do like i do i
think i do probabilistic inference maybe
sometimes
um but do science possess
the perspective of an external observer
uh this is another
question that the authors raise what

about organisms or brains like are the
are we
observers or not so what do you think
daniel do you have any any uh
ideas on these i think it's the perfect
paper to read as a starting point for
the discussion
because it really does depend on whether
we're a realist
or an instrumentalist because let's just
think about
computationalism like the idea that the
brain is anything like a computer or a
calculator has like anything to do with
numbers just the idea of
even talking about math in relationship
to the brain
if it's realism like the brain is a
computer versus instrumentalism it's
just a lens that we're using
to apply onto the um brain
it makes all the difference in the world
because it's the difference between
investigating the world and trying to
decipher what
it is quote versus knowing how to work
with the world
and then actually focusing our attention
on the integration of our preferences
and affordances with our models so it's
like a totally different mode of doing
research it's almost like in entomology
it's like the difference between doing
the basic research
going out there in the desert and
observing the ants versus like pest
control or integrated pest management
it's like you do need to know about the

basic biology to be effective
so it's not just as simple as observe
carefully and
think carefully it's like thinking
carefully and observing the system are
intrinsic but when there's this
integrated pest management mindset
then there's a different timeline
there's a different approach different
way of thinking
you might make a model at a different
level of agency and just say yeah
philosophy aside i'm going to model it
as if the group of colonies is related
i'm not making a claim about how they
are i'm just saying this is how i have
to model it because of xyz
constraint so just like the
instrumentalism
clarifies some philosophical debates on
these cognitive sciences
but it brings us right to the doorstep
of other questions
like our own unit of cognition our roles
and cognitive networks and things like
that
so it's a suite of questions that
i like it
clarified with the realism
instrumentalism and it just really
shows again how many uh downstream
consequences there are of taking a
research agenda that's realist versus
instrumentalist
but this time it's cognitive and that
means it really matters
so in the in my like background being
neuroscience i really think

fundamentally like we have to understand the tools that we are working with in order to be able to do good science like you know coming from a bench science aspect like you have to know the detectable limits of your spectrophotometer if you want to measure the concentration of protein right in a sample so it's like you need to know these limits of of your machine and and what exactly your machine is capable of and also what it's not capable of because you could be you know generating data with your instrument but the data is not not valid for one reason or another outside the limits of detection or whatever so um i think that understanding the brain and what the brain does and how it works is fundamental to doing good science because that's ultimately our instrument of detection very cool nice to bring it to the brain as both the instrument of our detection and really what it is like it is both and that's why and we're it and related to it or something so that's why it's kind of at this strange loop intersection let's go to the cognition in general slide so the working definition that you added was cognition is the mental action or process of acquiring knowledge and understanding

through thought experience in the census
and what were your
questions or kind of prompts for those
who would want to come on and
participate or what was exciting about
how the paper
rose these questions up for you
just the whole idea of
you know this the from again the like
the realism
aspects right like the is there
some way or or is it even possible
is cognition done in this bayesian
framework um i know that there are ways
to artificially
implement this framework in in a system
to
you know like an artificial intelligence
or some other thing for example but
but is do we have any of this actually
going on and i just wonder what what
other people think you know what where
after reading this paper that's so
heavily um
locked us into the instrumentalist uh
stance or perspective i just would like
to hear people's counterpoints
to that like i can see why the bayesian
framework is attractive
for cognition i can see that we
definitely um
but but not always like we don't always
learn from our mistakes
we tend to like repeat the same mistake
again and again and so it's not
always there's not always we don't
always update the the prior
right so so um you know we might miss a

data point or six

or twelve um and i just wanted to know

like what people's general thoughts are

on on whether cognition cognition is

bayesian

what that would mean and if bayesian you

know

then is it anything that we can ever

model mathematically

statistically is that a possibility

um yeah and then it's cool because the

first question is cognition bayesian

what's the alternative frequentism or

some other statistical

and then it's like whoa what if it

weren't statistical what if there was

something even beyond that clade

like that's an alternative that's local

but and then how do we even pull back

that far and you really

raised it with this piece here which is

that language and formal logic whether

philosophical logics or mathematical

formulas or any other type of

quantitative model

it's powerful for communication

especially in our current

niche but that's us today in this moment

and you wrote not everyone has an

internal monologue and i mean it's like

not

every species communicates the same way

and different people have different

experiences so

how will we tap into all of these

what if the logical is only the tip of

the iceberg or just the total

disinfo relative to the totality of what

needs to be understood
for effective action and that it's even
local optimization to be in the
linguistic
or the formal domain or it's predicated
deep on assumptions
that potentially are unwarranted or
inaccurate themselves
yeah definitely and i think that the
authors mentioned that also like or
maybe
uh not like language specifically but
they mentioned you know scientists
is is this the gloss that we're or is
this like actually the meat
that of what we're working with i don't
remember like the exact line from the
paper but i'll look it up for next time
the theory is called the bayesian brain
not the i use bayesian statistics our
package
to study the brain but that's what it
would be in the instrumentalist and i
think that
for people will say it's for reasons of
clarity we'll say well
the system's doing this or this is what
this protein wants to do it's really
easy
to confuse the map with the territory
to go from i'm using bayesian equations
or formats that remind me of basic
statistics or something i learned in a
bayesian course
i'm using that to model data it's the
bayesian brain
but again we didn't have the linear
regression brain when it was a linear

regression

and in some senses that's even what dcm

has underlying it as well

and so it's really something to think

about about

yeah what are the alternatives here and

how do we model it

yeah and so the authors like really

provide a fresh uh

i thought it was refreshing anyway

because people do they take the

you know it's the bayesian brain right

like this is how the brain works it's

basing of course

um and you know it's just like our

tendency to be so

anthropocentric i think like we do it

even within our our scientific

you know niche right we take what we're

studying or what we're focused on and

kind of

make it the center of everything i mean

the brain is like in my opinion

at like the center of our universe it is

the instrument by which we detect what

is going on

in the world so i mean just even

at the fundamental individual unit level

so so if if we're gonna say like oh this

is my work and now it's

the brain this is how the brain works so

i think that the brain has been

you know dissected and and re you know

imposed on so many framework frameworks

like going back to like broca's area and

the different like

phenomenology you know positions in the

brain and so everyone has kind of

taken their own their own thing and
tried to say this is how it is and the
way that the world works and
you know maybe it's not as simple as
that
agreed but it's just such a uh more
helpful framing to say well my
model the world versus your model of the
world and then my values are abc in that
order
and you explain this fraction of the
variance with this input here's your
values and let's figure out what model
is going to be
tractable for all of us versus world
view versus world view
it's um even without going down the
gerdell incompleteness
rabbit hole worldview versus worldview
is a difficult discussion to have
because you're talking past one another
with vocabulary
with the narrative with so many other
aspects but modeling is a shared
framework
now the question is does that
artificially
or unduly constrain the discussion
in a way where uh leaving the world view
level who really knows these are like
some of the coolest things so
i think let's go to this oh we have a
few more slides so let's kind of go
through
like that that's a perfect segue right
to the next um
slide right right about uh and this is i
i

put this in here um this is the brain
versus the mind right and i think that
the authors here
use that term at least in this place
interchangeably
and i'll just read this really quick um
one must be on the garden with respect
to organisms or brains unlike scientists
do not possess a perspective of an
external observer
thinking that it does requires a sound
argument that is not yet offered in the
literature in fact the hypothesis for
the brain as an ideal observer
has been often rejected in the
literature most recently
uh brett 2019 hippolyto at owl 2020 as a
bad metaphor
uh and then another group of
citations mursky and brickhard week
2019.
in agreement with this merski at owl
2020 called for encultured minds in
replacement
of error reduction minds oh and and that
like just
totally ties back to what you were just
saying you know the the different world
view like the encultured mind
it's got to be from which culture and
and you know can we speak the same
language if we're not both acculturated
in the same way
and also the encultured mind can be
carrying out
error reduction or surprise residual
reduction
but error reduction as the number one

leading point is like
well who is anyone to know what the gold
standard is
whereas when you start within cultured
minds like what are mines well they're
in culture it's developmental it's very
specific
what are they doing well among other
things they're
balancing error and surprise not down to
zero
but they're balancing it because they're
uncultured in a certain way
so we've started from how things kind of
actually are so it's almost starting
with
realism or naivety and then just asking
whether the
instrument can cover the space rather
than thinking we're going to have the
realism
and like the scope to describe how it
really is across space and
some cultures really are this way and
other cultures are really that way
versus i made a model and this parameter
ranges from
12 to 50. so continue
and then i just pulled these different
definitions out of mind is
is defined as i think google or someone
uh
the element of a person that enables
them to be aware of their world and
their experiences
to think and to feel the faculty of
consciousness
and thought and then the brain is an

organ of soft nervous tissue contained
in the skull of animals functioning as
the coordinating center of sensation
and intellectual and nervous activity um
and and i would be curious to hear
whether people think that these are the
same thing
are they related is the brain where the
mind is
um and can the are the is it okay to use
these two terms
interchangeably um and what is yeah i
think
we talked about error reducing mind
versus you know injured
are they one kind of thing monism are
they two kinds of things
dualism is there a qualification of
monism or dualism
um i also remember just in the past few
days forget where
heard something like the roots in the
mind and the fruits in the brain
or it was vice versa and i thought well
it could go either way
because that's really the debate is
whether matter generates mind or whether
mind
generates matter but blue another about
oracle determination but another day
let's go to this kind of closing
section so it's written here the fvp is
thus
best seen as a research heuristic a
particular lens through which we can
view and carve up the world
associated process theories is like
we'll make our joints as we go

associated process theories then are concerned with how the fep is realized in real world

systems this crucial distinction sets us to realize that the fvp

the fep does not imply process aspects or features

tandem duplication paralog such as representations

pertaining to the theoretical processes that aim to explain how the principle is realized

yet the fep does not in itself imply the representational tools implied by these process theories

so again there's a distinction between the principle and the process theories

so the instrument shop and the instruments themselves that's like the principle

to the process theories and then again the representational dimension is take it or leave it

and it turns out that neither form of representationalism

is gonna save realism and whatever you think about representationalism

as long as you're on the instrumentalist side it's all good and there's a debate to be had

and then at the bottom quote there is within this more modest framework of fep still plenty of insight to be gained into the workings of life and cognition by way of dynamic causal modeling

which is why we spent the time to go through it and why it's worth it to read

the textbook

because if this paper is on the right

track

then take it seriously what it says

which is that dynamic causal modeling is

underlying the fep

so metaphors you plow through them and

you're going to get to something that

has to be modeled like the dcm

in sum we argue that the modesty and

ambition go hand in hand

when it comes to models and the fep

awesome

great closing so blue any thoughts on

this or

what are these insights that we can gain

into life

and uh mind and brain cognition and

what else are you curious or excited

about

um so i'm curious and excited to hear

really what what people think

um about the mind the brain cognition

is cognition done in the mind is

cognition done in the brain

um where does that happen where does an

activism fit into this

kind of framework um is another like

parting question

um and so i'm excited really to hear

what people have to say

in the dialogue i could talk about what

i think all day long but i really would

would like to listen to what others have

to say

i agree also getting into the 16.0

um i think this is a fabulous lead up to

discussing consciousness

uh in 16.0 agreed it's all part of a
sequence i would say i'm excited for
both the authors to join for both weeks
and also for
hopefully new and old participants to
join
and just to hear everyone's perspective
like
before i read the paper i felt this way
then when i was reading the paper i
thought i stopped here
and i took a break here and i took a
week off here i thought about this part
and
i cut out this part and i highlighted
this part like what this question arose
just
what was the before what happened during
working through the paper
if you had 20 minutes before the meeting
it's chill you can still read a few
sentences something will still pop out
and then
we capture that as it's happening with
our discussions and we bring out things
that people can't bring out
alone or in dialogue with the paper
itself
and then it's like how we're going to
move forward and so how does it change
how we work with the fep and active
inference how we communicate it and do
it
to be instrumentalist how do we be
instrumentalist through and through
so that we don't put out the realist
bait and tell people the fep is a grand
unifying theory of how things are

okay well that's a realist claim so how
do we start with the instrumentalist
approach
in our minds understanding this paper
and conversing with the authors and
hearing from those who are just learning
at all different stages and then
imbue that into our full stack approach
to making tools onboarding and
accessibility just all these other
aspects of what we do
related to fep and active inference what
are the implications
for instrumentalism in those cases and
that will be pretty fun to find out so
oh oh then let's close with the tragedy
because this is the uh this is the the
danume so i control f'd
but i didn't find tragedy in the paper
so thinking
it's you know it's the dog that doesn't
bark in the night
because it's the tragedy is the title of
the paper but what is the tragedy
and so this quote they write the
difference between realism and
instrumentalism
is primarily ontological in nature in
realism there's an ontological claim
with regards to the status of models and
living systems
in instrumentalism there is no such
ontological claim about how things are
ontology
this may be seen as a weakness as though
it looks as though the instrumentalist
position only gives up explanatory
ambitions

relative to the fep realist so in other words is the instrumentalist just picking a lower target a easier target and so like yeah we did it but it's like too easy it's not useful actually we need real we need full-blown realism to be effective this is true devastating true however the ambitions given up on we argue are never going to be met so it was a delusion for the first thought or for those who were aiming it's great you know shoot for the moon end up in the stars or whatever but that was not the right target if this is going on the right track the realist ambition is a fata morgana if you will as such instead of chasing ghosts the instrumentalist position is more realistic in their ambition and so that was pretty fun i had to look up what it was but it's this visual illusion on the horizon over water or over the desert that does all different kinds of visual effects blue you probably see it in the southwest in the desert when you're driving and stuff could look like a boat or a car or a river or mercury and so it's kind of interesting it's like we're in a desert you know there's a real niche that is challenging either there's relative con competition

within the niche
or all the agents in the niche just get
wiped off the table
so we're in a scenario that involves
doing inference
and it might feel good to have a
delusional target
but in the end we need policy that works
and keeps us alive and so we don't want
to get tied up with
illusions of what things could be we
want to be
on a path towards utility but also
bringing in all these other nuanced
characteristics so it's a tough question
and it's a beautiful
illusion and an allegory and so i was
just wondering what is the tragedy or
is realism a tragedy or what is the
tragedy in this case
blue before we look at the definition of
tragedy
in this case i think that the tragedy is
that uh representationalism
is is not even applicable in the end
right
so and that's my tragedy because i'm so
attached to my
you know representationalist uh
standpoint i guess let's go to what the
final slide here is just
the definition of tragedy we won't read
it but
it was just pretty interesting to see
that uh tragedy has meant different
things
over time of course it's been used in
different ways but it's sort of like

an inciting incident incident a plot
reversal
and then there's the end and then
there's a catharsis and a new resolution
melt thaw resolve itself into a do
except in this case it's d
o because it's doing stuff so it's sort
of like the prior
and then the update or the stimuli or
the observation
and then there's something the
resolution
could be the same as the beginning or
the person could have resolved and
everyone could have
you know gone home with a different you
know set of whatever
but it has this format that's the
minimal
problem reaction solution or prior
update posterior so pretty interesting
to see how tragedy has a connotation
of like sad it was a tragedy what
happened it's like oh something sad must
have happened
but this is almost neutral and it's like
saying oh it's a tragedy what happened
it's like
what happened what was the update what
was the plot reversal that led to the
outcome so
very nuanced reference and very
interesting by
the authors so here are the final
questions blue any last thoughts
on anything otherwise we'll close it out
yeah i think that's good i'm excited for
the dialogue thanks so much for

joining for participating being the live
participant

everyone's welcome to follow up
add comments on any platform or
get in touch with us it's just gonna be
a good conversation

blew it thanks again really special to
have our first zero dialogue and
everyone's welcome on dot zero
everyone's welcome on

1.2 just get in touch if you want to
participate

and see you later bye

bye thanks